

SOLVAER III, NorwayWMO: **011210**Lat: **66.371N**Lon: **12.611E**Elev: **10**StdP: **101.20**Time Zone: **1.00 (EUC)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.2	-5.7	-13.9	1.1	-5.2	-12.5	1.3	-4.5	19.4	-0.1	17.9	-1.6	12.2	120	0.852

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	21.2	15.3	19.2	14.1	17.6	13.3	16.1	20.0	14.9	17.9	13.9	16.6	5.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	10.1	17.3	13.3	9.5	16.3	12.5	9.0	15.4	45.0	20.2	41.6	18.2	39.2	16.6	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.9	13.4	DB	-8.2	24.7	2.4	2.3	-9.9	26.4	-11.3	27.7	-12.7	29.0	-14.4	30.7	(4)
(5)			WB	-9.6	17.5	1.9	1.8	-11.0	18.8	-12.1	19.8	-13.1	20.8	-14.5	22.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	0.7	0.7	1.6	4.6	8.1	10.5	13.3	13.6	11.5	7.4	4.0	2.2		
	DBStd	5.44	3.11	3.26	2.69	2.65	3.08	2.73	2.89	2.33	2.32	2.59	2.95	3.04		
	HDD10.0	1593	288	261	261	164	77	24	1	0	11	86	179	241		
	HDD18.3	4305	547	494	519	411	318	236	159	148	206	338	428	499		
	CDD10.0	334	0	0	0	3	17	38	104	111	54	7	0	0		
	CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0	0		
	CDH23.3	21	0	0	0	0	1	1	18	2	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0		
(14) Wind	WSAvg	6.9	8.8	8.6	7.6	6.2	5.9	5.9	5.3	5.1	5.8	7.3	7.7	8.3		
(15) Precipitation	PrecAvg	1717	163	150	148	111	100	96	99	114	185	198	160	193		
	PrecMax	2301	313	306	293	238	158	209	245	291	489	419	331	372		
	PrecMin	1263	0	31	27	26	31	5	25	8	25	43	15	4		
	PrecStd	292	84	72	66	57	37	57	65	63	110	101	94	97		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	7.0	8.3	15.3	19.6	20.7	25.6	22.7	18.7	15.2	11.2	8.5		
		MCWB	5.9	5.2	4.5	8.8	12.1	13.8	18.1	16.6	13.4	10.6	8.8	6.4		
	2%	DB	6.5	6.3	6.7	12.7	16.7	18.4	22.5	20.3	17.2	13.5	9.7	7.7		
		MCWB	4.6	4.8	4.8	7.7	10.8	12.7	16.6	15.0	12.5	9.5	7.6	6.3		
	5%	DB	5.7	5.7	6.2	9.4	14.6	16.1	19.6	18.7	15.9	12.1	8.7	7.1		
		MCWB	4.0	4.2	4.4	5.9	10.1	11.7	15.0	14.1	11.6	9.0	6.6	5.5		
	10%	DB	4.7	4.9	5.5	7.8	12.3	14.2	17.2	17.3	14.8	10.7	7.8	6.2		
		MCWB	3.1	3.3	3.6	5.1	8.6	11.1	13.9	13.6	11.2	8.5	5.7	4.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	6.2	6.4	9.0	12.9	14.8	19.0	17.2	14.1	11.5	9.6	7.7		
		MCDB	7.3	6.5	6.7	14.9	19.1	19.7	24.7	21.7	17.2	14.6	10.5	8.0		
	2%	WB	5.3	5.5	5.7	7.8	11.4	13.2	16.8	15.6	13.1	10.3	8.0	6.9		
		MCDB	5.9	6.1	6.4	12.0	15.7	17.3	21.4	19.2	15.7	12.5	9.2	7.6		
	5%	WB	4.2	4.5	4.7	6.5	10.2	12.1	15.5	14.6	12.4	9.5	6.9	5.7		
		MCDB	5.3	5.5	6.0	8.6	14.1	15.0	18.7	17.8	14.8	11.4	8.7	6.7		
	10%	WB	3.2	3.4	3.7	5.7	8.9	11.4	14.4	13.8	11.8	8.7	6.0	4.7		
		MCDB	4.6	4.6	5.3	7.2	11.7	13.8	16.7	16.6	14.1	10.6	7.5	6.1		
(35) Mean Daily Temperature Range	MDBR		2.8	2.9	3.2	3.7	4.2	3.6	4.1	4.1	3.5	2.8	2.8	2.9		
	5% DB	MCDBR	3.4	3.6	3.7	5.5	8.1	7.3	8.7	7.1	5.0	3.6	3.1	3.7		
		MCWBR	3.4	3.3	3.2	3.0	3.8	3.3	3.9	3.2	2.6	2.4	2.6	3.5		
	5% WB	MCDBR	3.4	3.6	3.9	4.8	7.5	6.2	7.9	6.7	4.5	3.2	3.2	3.7		
		MCWBR	3.5	3.6	3.8	2.9	4.0	3.1	3.9	3.3	2.7	2.9	3.2	3.7		
(40) Clear-Sky Solar Irradiance	taub		0.369	0.300	0.327	0.350	0.350	0.346	0.355	0.342	0.331	0.306	0.348	0.366		
	taud		2.539	2.182	2.282	2.361	2.443	2.500	2.500	2.525	2.521	2.323	2.500	2.489		
	Ebn at Noon		195	539	727	810	852	865	844	819	739	538	182	31		
	Edh at Noon		19	52	78	93	94	92	89	79	63	45	17	7		
(44) All-Sky Solar Radiation	RadAvg		0.06	0.41	1.59	3.19	4.48	5.04	4.53	3.43	1.85	0.70	0.12	0.01		
	RadStd		0.01	0.08	0.24	0.33	0.31	0.65	0.49	0.36	0.24	0.14	0.01	0.00		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-93	-133	N/A

Nomenclature: See separate page